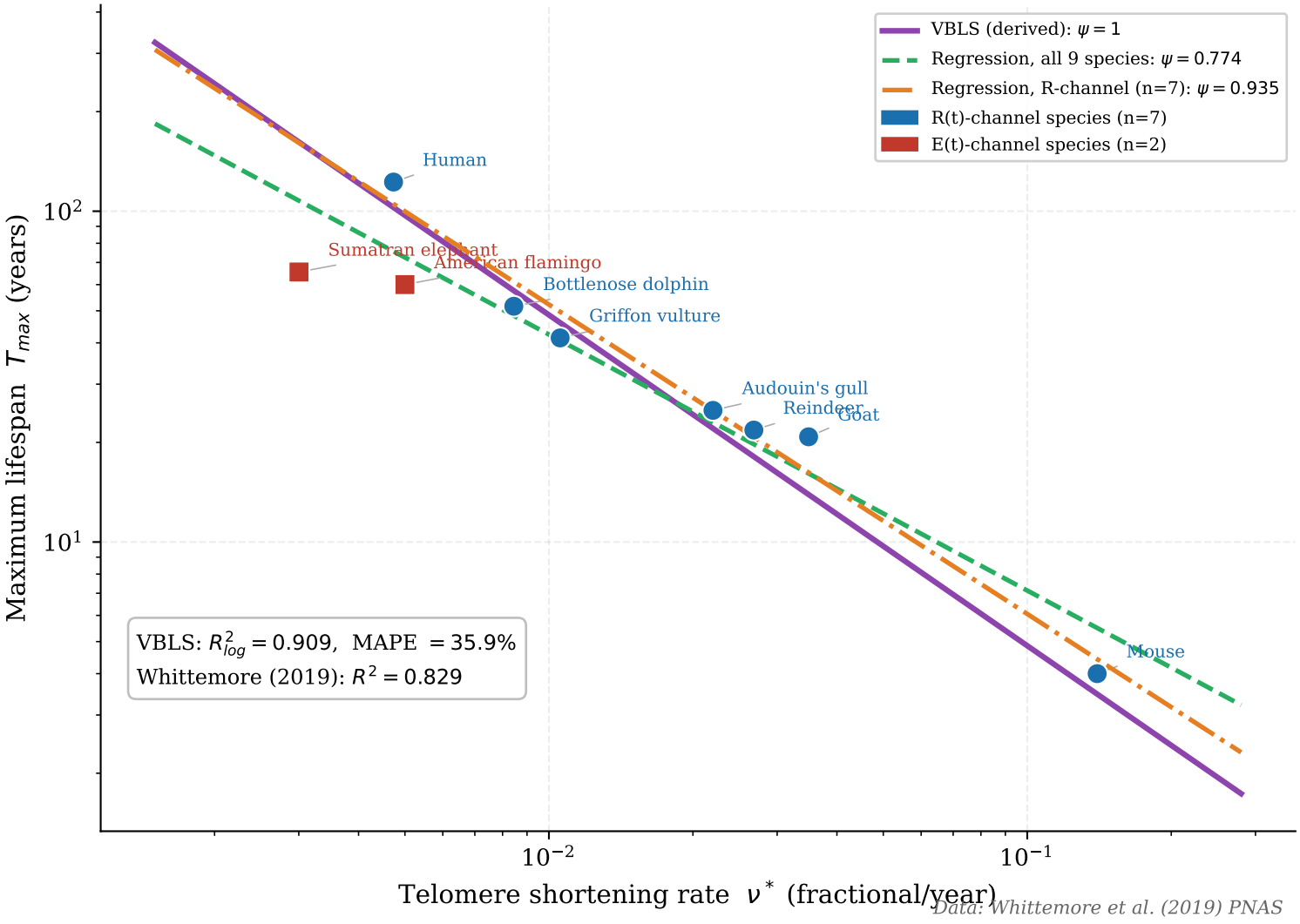
Power Law $T_{max} \sim \nu^{*-1}$: VBLS Derivation vs Empirical Regression
VBLS derives $\psi = 1$ analytically (Variant A, all 9 species); regression fits post hoc
VBLS (derived): $\psi = 1$
Regression, all 9 species: $\psi = 0.774$
Regression, R-channel (n=7): $\psi = 0.935$
R(t)-channel species (n=7)
E(t)-channel species (n=2)
Human
Sumatran elephant
American flamingo
Bottlenose dolphin
Griffon vulture
Audouin's gull
Reindeer
Goat
Mouse
VBLS: $R^2_{log} = 0.909$, MAPE = 35.9%
Whittemore (2019): $R^2 = 0.829$
Maximum lifespan T_{max} (years)
10^2
10^1
10^{-2}
10^{-1}
Telomere shortening rate ν^* (fractional/year)
Data: Whittemore et al. (2019) PNAS